Cosine-Normalized Classifiers in Federated Continual Learning

IEEE-style report with methodology, mathematical formulation, and results

Abstract— We study a federated continual learning (FCL) pipeline based on Accurate Forgetting (AF-FCL) and compare the baseline linear classifier against a cosine-normalized classifier. The report explains the full pipeline from sequential task learning and replay generation to the classifier change, then summarizes the reported Non-IID and Shared-split results, showing that cosine normalization improves accuracy and stabilizes flow training.

I. Introduction

Federated continual learning learns tasks one after another across clients without keeping raw historical data. This creates two main issues: catastrophic forgetting and instability under heterogeneous data. The baseline AF-FCL pipeline addresses forgetting using replay generated by a conditional normalizing flow and a distillation loss, while our modification replaces the standard linear classifier with a cosine-normalized classifier to reduce magnitude bias and improve angular separation between classes. The reported results in the source document are summarized and reformatted in this paper-style report. ■filecite■turn0file0■

II. From-Scratch Methodology

A. Baseline AF-FCL pipeline

The baseline model uses a feature extractor, a classifier head, and a conditional flow module. For an input feature vector x, the standard classifier computes class logits z_c = W_c^T x + b_c and applies a softmax over classes. Since old client data cannot be retained, the flow model learns a feature distribution p(x | y) so that replay features x~ can be sampled for earlier classes. Training combines current-task cross-entropy, replay cross-entropy, and knowledge distillation.

B. Why the linear classifier is limited

The linear score W^T x depends on both angle and magnitude, because W^T x = ||W|| ||x|| cos(theta). During continual updates, a class can gain a larger norm simply because it was updated more recently or more aggressively. This introduces magnitude bias: a class with a larger norm may dominate even if the angular match is weaker. In FCL, that bias can contribute to forgetting and class imbalance.

C. Cosine-normalized classifier

We replace the classifier score with z_c = s * (W_c^T x) / (||W_c|| ||x||), where s is a learnable or fixed temperature scale. After normalization, the score becomes z_c = s cos(theta_c), so the decision depends only on angular similarity. This means that two vectors pointing in the same direction receive the same score regardless of their lengths. That is the key reason cosine classifiers often work well in class-incremental and continual-learning settings.

D. Loss function

The overall learning objective remains the AF-FCL structure: L = L_CE(current) + L_CE(replay) + L_KD. The only change is the classifier geometry used to produce logits. In practice, this means we still learn new tasks, replay old tasks, and distill previous knowledge, but with angular scoring instead of raw linear scoring.

III. Mathematical Formulation

For each class c, the baseline and cosine classifier scores are:

Baseline: z_c = W_c^T x + b_c

Cosine: z_c = s · (W_c^T x)/(||W_c|| ||x||) = s · cos(theta_c)

If W_c and x are normalized as W■_c = W_c / ||W_c|| and x■ = x / ||x||, then z_c = s (W■_c^T x■). Because the dot product of two unit vectors equals cosine similarity, the classifier becomes angle-based. This makes the classifier less sensitive to feature scale drift and class-norm drift across tasks.

IV. Results

A. Non-IID class-so-far accuracy

Task	Baseline-CPU	Baseline-GPU	Cosine (80ep)
t=1	27.11	26.73	62.80
t=2	33.94	34.16	49.16
t=3	39.28	49.11	46.75
t=4	41.04	40.84	33.49
t=5	42.46	35.01	35.80
t=6	36.12	35.27	42.73

The cosine classifier gives a strong early boost on the Non-IID split and finishes above the baseline at the final task. The strongest improvement is at t=1, where the cosine model reaches 62.80% versus about 27% for the baseline.

B. Non-IID final per-task accuracy

Task	Baseline-CPU	Baseline-GPU	Cosine
T1	27.52	24.45	35.48
T2	15.30	12.53	17.81
T3	38.30	38.70	40.13
T4	29.34	29.67	33.69
T5	38.90	39.48	44.02
T6	67.34	66.81	85.25

The final-task result is especially important for continual learning because it reflects retention after the whole sequence. The cosine model reaches 85.25% on T6, a large gain over the baseline runs.

C. Non-IID flow-loss stability

Configuration	Median	Min	Max
Baseline-CPU	96,449	518	1.00 × 10^18
Baseline-GPU	20,998	521	1.53 × 10^18
Cosine (80ep)	518	470	8.38 × 10^11

Lower flow loss indicates better numerical stability. The cosine configuration is dramatically more stable than the baseline according to the reported median values.

D. Shared-split accuracy

Task	Baseline	Cosine (30ep)	Cosine (50ep)	Cosine (150ep)
t=1	55.95	67.27	70.43	75.58
t=2	54.70	71.85	74.00	75.67
t=3	49.66	75.06	73.66	74.96
t=4	49.50	69.65	70.89	62.02
t=5	52.87	75.69	72.36	70.76
t=6	61.27	87.44	86.41	84.40

On the Shared split, cosine normalization is consistently better than the baseline across all tasks and epochs, with the best final class-so-far score of 87.44% at 30 epochs.

E. Shared final per-task accuracy

Task	Baseline	Cosine (30ep)	Cosine (50ep)	Cosine (150ep)
T1	60.55	85.62	84.73	89.02
T2	62.98	85.70	86.49	85.70
T3	54.39	84.98	84.65	79.56
T4	63.02	91.77	89.22	84.49
T5	54.86	86.03	81.75	75.35
T6	71.83	90.52	91.59	92.24

The cosine classifier improves all task-wise final accuracies. In particular, T6 rises from 71.83% in the baseline to 92.24% at 150 epochs.

F. Shared flow-loss stability

Configuration	Median	Min	Max
Baseline	28,209	514	1.71 × 10^17
Cosine (30ep)	470	470	5.26 × 10^2
Cosine (50ep)	470	470	1.42 × 10^5
Cosine (150ep)	642	470	1.77 × 10^13

Again, the cosine head shows markedly improved numerical stability compared with the baseline.

V. Discussion

The key takeaway is that the cosine head changes only the classifier geometry, not the whole AF-FCL pipeline. Replay generation, distillation, and sequential training remain intact. Yet this small geometric change significantly improves accuracy and stability because classification decisions are made on angular similarity rather than raw vector norms. This is especially useful in FCL, where task shifts can inflate or shrink norms in a way that harms older classes.

VI. Conclusion

The redesigned classifier produces a cleaner and more robust decision boundary in federated continual learning. In the reported experiments, cosine normalization improves accuracy on both Non-IID and Shared splits and reduces flow instability. That makes it a strong practical upgrade over the plain linear classifier in the AF-FCL framework. ■filecite■turn0file0■